

Ecopath with Ecosim technical papers: Spatial Temporal Data Framework Configuration file layout



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Abstract

With the introduction of the Spatial Temporal Data Framework (e.g., *framework*), Ecospace, the spatial-temporal module of the EwE modelling approach, gained the possibility to vary its input maps over time via time series of maps that are available outside an EwE model. The framework is described in detail in Steenbeek (2012) and Steenbeek et al (2013).

There are many potential external spatial temporal data sources of interest to EwE applications. Spatial Temporal Data Framework Configuration files (e.g., *configuration files*) give EwE modellers the possibility to bundle a select number of external data sources to be used for a specific modelling application. Although the EwE desktop software maintains the contents and integrity of configuration files, EwE modellers and programmers may wish to dynamically generate their own configuration files especially when running the EwE software in scripted environments without user interface.

This document describes the structure of the configuration file to assist in such efforts.

Keywords

Ecopath with Ecosim, EwE, Ecospace, Spatial Temporal Data Framework, GIS

Credits

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Version history

Date	Version	Author	Description
2015-01-14	1.0	Jeroen Steenbeek	Initial version

Configuration file

Ecospace is able to find this external data through configuration files. A configuration file is a XML file that defines one or more available external series of maps over time (e.g., *datasets*). Each dataset is a collection of time-stamped maps that can be used in Ecospace to drive the content of a specific Ecospace driver map.

Base structure

The XML configuration file has the following base structure:

```
<?xml version="1.0" standalone="yes"?>
<Datasets Name=".." Author=".." Contact="" Station=".." Description="..">
  <Dataset Type=".." GUID="..">
    <Configuration>
      ..
    </Configuration>
  </Dataset>
</Datasets>
```

Listed nodes have the following cardinality:

- The `<?xml>` node must occur first, must occur exactly once, and is mandatory
- The `<datasets>` node is mandatory and must occur exactly once
- The `<datasets>` node can have zero or more nested `<dataset>` nodes
- Each `<dataset>` must have exactly one nested Configuration node

Listed XML attributes have the following meaning:

Field	Format	Description
Name	Text	Human-legible name describing the content of the configuration file
Author	Text	The author of the configuration file
Contact	Text	The contact information of the author
Station	Text	The work station that the configuration file was created on
Description	Text	Human-legible description of the content of the configuration file
Type	Text	The type class name of the dataset, via which the framework locates the correct dataset class in the EwE source code to handle the following configuration section. Available type class names are described later in this document
GUID	Text	A globally unique identifier or <i>GUID</i> for the dataset, assigned by EwE. The EwE model database uses this ID to persistently connect datasets to Ecospace driver layers. The GUID format is described in detail http://en.wikipedia.org/wiki/Globally_unique_identifier .

For each dataset, the configuration file holds the following information:

- Dataset type and unique identification number
- Dataset name and description
- Dataset origin and authorship
- Definition(s) for temporal maps, which encompasses the following for each map:
 - A time stamp so that Ecospace can temporally align each map during execution
 - A reference where the actual map is located

- Optionally, the spatial bounding box for a map (see notes, at the end of this document)

Dataset types

Currently, the EwESpatialAssetsPlugin defines four types of dataset classes:

- Single file dataset
- Multi-file dataset
- ASCII file dataset
- CSV file dataset

Single file dataset

The single file dataset provides access to exactly one map file in a common GIS file format.

Class name

The Dataset type class name for a single file dataset is:

```
EwESpatialAssetsPlugin!EwESpatialAssetsPlugin.SpatialData.cSingleFileDataSetPlugin
```

Configuration XML

The <configuration> XML for this class of dataset has the following structure:

```
<Configuration>
  <Name></Name>
  <Description></Description>
  <Variable></Variable>
  <File IsRelative=".." Date=".."></File>
</Configuration>
```

Child nodes of <configuration> for the single file dataset have the following cardinality:

- The <name> node is mandatory, and must occur exactly once
- The <description> node is optional, and can occur only once
- The <variable> node is optional, and can occur only once
- The <file> node is mandatory, and must occur exactly once

Listed XML nodes and attributes have the following meaning:

Node/attribute	Format	Description
Name	Text	Human-legible name for the dataset
Description	Text	Human-legible description for the dataset
Variable	Text	A EwE variable name, populated and used only by the EwE user interface for filtering content. The variable name is directly derived from eVarName enumerated values in the EwE source code
File	Text	File name of the map for this dataset. This file name can be an absolute path, or a path relative to the location of the configuration file
IsRelative	Boolean	Flag that indicates if the file name indicates a relative path. (True/False)
Date	Date	The file can have a time stamp (yyyy-mm-dd) or can be without time. If no time stamp is given, the spatial temporal data framework will apply the file at the very first Ecospace time step.

Example

The following XML configuration file snippet shows a single file dataset definition:

```
<Dataset
Type="EwESpatialAssetsPlugin!EwESpatialAssetsPlugin.SpatialData.cSingleFileDataSetPlugin"
GUID="">
  <Configuration>
    <Name>RelPP_025dd</Name>
    <Description>Relative PP at 0.25 decimal degree, Adriatic</Description>
    <Variable>NotSet</Variable>
    <File IsRelative="True" Date="2010-01-01">Adriatic\RelPP-2010-01-01.asc</File>
  </Configuration>
</Dataset>
```

Multi-file dataset

The multi-file dataset provides access to a time series map files. All files must of the same common GIS file format.

Class name

The class name that should be used in the Dataset Type attribute is:

```
EwESpatialAssetsPlugin!EwESpatialAssetsPlugin.SpatialData.cMultiFileDataSetPlugin
```

Configuration XML

The <configuration> XML has the following structure:

```
<Configuration>
  <Name></Name>
  <Description></Description>
  <Variable></Variable>
  <Source IsRelative=".."></Source>
  <Seasonal Date=".."></Seasonal>
  <Files>
    <File Date=".."></File>
    ..
  </Files>
</Configuration>
```

Child nodes of <configuration> for the multi-file dataset have the following cardinality:

- The <name> node is mandatory, and must occur exactly once
- The <description> node is optional, and can occur only once
- The <variable> node is optional, and can occur only once
- The <source> node is mandatory, and must occur exactly once
- The <seasonal> node is optional, and can occur only once
- The <files> node is mandatory, and can occur only once
- The <files> must contain one or more <file> nodes

Listed XML nodes and attributes have the following meaning:

Node/attribute	Format	Description
Name	Text	Human-legible name for the dataset
Description	Text	Human-legible description for the dataset
Variable	Text	A EwE variable name, populated and used only by the EwE user interface for filtering content. The variable name is directly derived from eVarName enumerated values in the EwE source code
Source	Text	Path to the folder with the maps for this dataset. The folder path can be an absolute path, or a path relative to the location of the configuration file
IsRelative	Boolean	Flag that indicates if the Source field indicates a relative or absolute path (True/False)
Seasonal	Boolean	States if the data repeats after one year (True/False)

EndDate	Date	The last date (yyyy-mm-dd) for seasonal data
Files	-	The one node that holds all files in the dataset
File	Text	The file name of a map in the dataset, without any path information
Date	Date	The time stamp (yyyy-mm-dd) for the map

Example

The following XML configuration file snippet shows a multiple file dataset definition:

```
<Dataset
Type="EwESpatialAssetsPlugin!EwESpatialAssetsPlugin.SpatialData.cMultiFileDataSetPlugin"
GUID="21437045-a4f0-4e7f-bfbb-ae3479f85dd8">
  <Configuration>
    <Name>AMIS PP (bgd)</Name>
    <Description>JRC AMIS PP</Description>
    <Variable>LayerRelPP</Variable>
    <Source IsRelative="False">G:\Data\Spatial data\PP\JRC_org\PP_1997-2008</Source>
    <Seasonal Date="9999-12-01">False</Seasonal>
    <Files>
      <File Date="1997-09-01"/>AMIS S PP 1997 09.bgd</File>
      <File Date="1997-10-01"/>AMIS_S_PP_1997_10.bgd</File>
      <File Date="1997-11-01"/>AMIS_S_PP_1997_11.bgd</File>
    </Files>
  </Configuration>
</Dataset>
```

ASCII file dataset

The ASCII file dataset is identical to the multi file dataset in all aspects except for two major differences:

- The dataset can only read ESRI ASCII files
- The ASCII files are presumed to exactly line up with an Ecospace scenario. Spatial bounding box, number of rows and columns, and cell size are presumed to perfectly line up with an Ecospace scenario.

This type of dataset will very bluntly copy its content to an Ecospace driver map: the top-left value in the ASCII grid will be placed in the top-left cell of an Ecospace map, the second ASCII cell into the second map cell, and so forth. This dataset performs no spatial checking, and omits all GIS manipulations.

Because all GIS operations are omitted when reading data, this type of dataset performs much faster than the true GeoSpatial dataset types. However, the user is entirely responsible for preparing the driver maps to match an Ecospace scenario.

Class name

The class name that should be used in the Dataset Type attribute is:

```
EwESpatialAssetsPlugin!EwESpatialAssetsPlugin.SpatialData.cASCIIFilesDataSetPlugin
```

Configuration XML

The <configuration> XML is identical to the multi-file dataset.

Example

The following XML configuration file snippet shows an ASCII file dataset definition:

```
<Dataset
Type="EwESpatialAssetsPlugin!EwESpatialAssetsPlugin.SpatialData.cASCIIFilesDataSetPlugin"
GUID="c2938c96-964e-414d-b641-f827db515169">
```

```

<Configuration>
  <Name>Georgia Strait RelPP 0.1dd</Name>
  <Description>Rel PP map prepared for Georgia Strait 0.1dd scenario</Description>
  <Variable>LayerRelPP</Variable>
  <Source IsRelative="True">RelPP-GeorgiaStraight</Source>
  <Seasonal EndDate="9999-12-01">False</Seasonal>
  <Files>
    <File Date="1979-01-01">RelPP 1979-01-01.asc</File>
    <File Date="1979-04-01">RelPP 1979-04-01.asc</File>
    <File Date="1979-07-01">RelPP 1979-07-01.asc</File>
    <File Date="1979-10-01">RelPP_1979-10-01.asc</File>
  </Files>
</Configuration>
</Dataset>

```

CSV file dataset

The CSV file dataset is identical to the multi file dataset in all aspects except for two major differences:

- The dataset can only read comma-separated files depicting row / column data, which contain sparse point data for a specific Ecospace driver map
- The CSV files can either indicate data by lat+lon, or by row+col

Class name

The class name that should be used in the Dataset Type attribute is:

```
EwESpatialAssetsPlugin!EwESpatialAssetsPlugin.SpatialData.cCSVFilesDataSetPlugin
```

Configuration XML

The <configuration> XML is identical to the multi-file dataset.

Example

The following XML configuration file snippet shows an CSV file dataset definition:

```

<Dataset
Type="EwESpatialAssetsPlugin!EwESpatialAssetsPlugin.SpatialData.cCSVFilesDataSetPlugin"
GUID="e0dbe785-cfce-4afe-988d-63cebc1426fa">
  <Configuration>
    <Name>A series of Ecospace [row, col, value] CSV files over time</Name>
    <Description>A series of CSV files in row, col, values* format, as exported by Ecospace.
    Note that these files lack any spatial reference, and are intended to be used only with the
    Ecospace scenario from which they were generated.</Description>
    <Variable>NotSet</Variable>
    <Source IsRelative="False">G:\Data\Spatial data\PP\Adriatic RelPP</Source>
    <Seasonal EndDate="9999-12-31">False</Seasonal>
    <Files>
      <File Date="1950-01-01">Adriatic bal75 initial-PP (relative).csv</File>
      <File Date="1950-02-01">Adriatic bal75 initial-PP (relative)2.csv</File>
    </Files>
  </Configuration>
</Dataset>

```

Notes

File nodes may contain the attributes Indexed, MinLat, MaxLat, MinLon, and MaxLon. These attributes are the result of indexing, which encompasses a simple spatial bounds check made by the framework for each GIS file.

References

- Steenbeek, J., 2012. Bridging the gap between ecosystem modelling tools using geographic information systems (MSc). Manchester Metropolitan University, Manchester.
- Steenbeek, J., Coll, M., Gurney, L., Mélin, F., Hoepffner, N., Buszowski, J., Christensen, V., 2013. Bridging the gap between ecosystem modelling tools using geographic information systems: driving a food-web model with spatial-temporal primary production data. *Ecol. Model.* 263, 139–151. doi:2013